








Advanced Time Series Analysis and Forecasting

Chapter 8: Advanced volatility modelling: realised measures, HAR and multivariate GARCH



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Acronyms used in this material (1/3)

- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARMA**: AutoRegressive Moving Average (model)
- **BEKK**: Baba–Engle–Kraft–Kroner (multivariate GARCH model)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNHLS**: Barndorff-Nielsen–Hansen–Lunde–Shephard
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BV**: Bipower Variation
- **CAC**: Cotation Assistée en Continu (CAC 40 index, Paris) — the CAC 40 index of the Paris bourse
- **CCC**: Constant Conditional Correlation (model)
- **CLT**: Central Limit Theorem
- **DCC**: Dynamic Conditional Correlation
- **DCC-NL**: DCC with Nonlinear shrinkage of the target (Engle–Ledoit–Wolf 2019)
- **DM**: Diebold–Mariano (test)
- **EGARCH**: Exponential GARCH
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **FDR**: False Discovery Rate

Acronyms used in this material (2/3)

- **FRED**: Federal Reserve Economic Data
- **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- **FWER**: Family-Wise Error Rate
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GARCH-MIDAS**: GARCH with a MIxed DAta Sampling long-run component (Engle–Ghysels–Sohn 2013)
- **GARCH-X**: GARCH with eXogenous variables
- **GJR**: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- **GMM**: Generalized Method of Moments
- **GMV**: Global Minimum-Variance (portfolio)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HAR-CJ**: HAR with Continuous and Jump components
- **HARQ**: HAR with realised Quarticity
- **HEAVY**: High-frEquency-bAsed VolatilitY (models)
- **IGARCH**: Integrated GARCH
- **IQ**: Integrated Quarticity
- **LLM**: Large Language Model
- **MAE**: Mean Absolute Error
- **MFM**: Modelling Financial Markets
- **MIDAS**: MIxed DAta Sampling (regression)
- **ML**: Machine Learning

Acronyms used in this material (3/3)

- **MSE**: Mean Squared Error
- **MSTR**: MicroStrategy (Strategy Inc.), NASDAQ ticker
- **OLS**: Ordinary Least Squares
- **OPG**: Outer Product of the Gradient (scores)
- **PPI**: Producer Price Index
- **QLIKE**: Quasi-LIKElihood loss
- **QML**: Quasi-Maximum Likelihood
- **QMLE**: Quasi-Maximum Likelihood Estimator
- **QV**: Quadratic Variation
- **RGARCH**: Realized GARCH
- **RK**: Realised Kernel
- **RMSE**: Root Mean Squared Error
- **RQ**: Realised Quarticity
- **RV**: Realised Variance
- **SV**: Stochastic Volatility
- **TSA**: Time Series Analysis
- **TSLA**: Tesla Inc., NASDAQ ticker
- **TSRV**: Two-Scales Realised Variance
- **US**: United States
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- **VEC**: VEC multivariate GARCH (half-vectorised covariance)
- **VIX**: CBOE Volatility IndeX
- **VR**: Variance Ratio

Today's question and route

- **Question**: how precisely can we measure, model and forecast a variance that is never observed?
  - and which of our tools still work when the data are noisy, jumpy or high-dimensional?
  - two routes to volatility: a parametric filter of daily returns (GARCH) and a nonparametric measurement from intraday prices (realised measures); the frontier combines them
- **Route** of the chapter
  - quasi-maximum likelihood for GARCH and robust inference; long-run components: component GARCH and GARCH-MIDAS with macroeconomic drivers
  - realised measures: asymptotics of realised variance, microstructure noise, realised kernels, bipower variation and jump tests
  - forecasting with realised measures: HAR, HARQ, Realized GARCH, HEAVY; robust loss functions
  - multivariate: BEKK, DCC and cDCC, the curse of dimensionality and DCC-NL, realised covariance
- We build on TSA, Chapter 5 (GARCH, GJR, EGARCH, news impact, QLIKE) and TSA, Chapter 14 (DCC basics)
  - MFM, Chapters 5–9 apply these tools to market risk; Seminar 8 comes before this lecture

Learning outcomes

- State the conditions under which Gaussian QML for GARCH is consistent and asymptotically normal, and compute Bollerslev–Wooldridge standard errors
- Separate short-run from long-run volatility with component GARCH and GARCH-MIDAS, and judge whether a macroeconomic driver matters
- Derive the limit theory of realised variance, explain the effect of noise and jumps, and choose among RV, realised kernels and bipower variation
- Forecast volatility with HAR, HARQ and Realized GARCH, and compare forecasts with losses that are robust to a noisy proxy
- Estimate DCC and cDCC models, explain why large covariance matrices need shrinkage, and measure correlation from intraday data

Reading, data and tools

- Backbone: Francq and Zakoïan (2019); Andersen, Bollerslev, Diebold and Labys (2003); Barndorff-Nielsen and Shephard (2002); Barndorff-Nielsen, Hansen, Lunde and Shephard (2008)
  - forecasting and the multivariate case: Corsi (2009); Hansen, Huang and Shek (2012); Engle (2002); Patton (2011)
  - case studies: Engle, Ghysels and Sohn (2013) (GARCH-MIDAS), Bollerslev, Patton and Quaedvlieg (2016) (HARQ), Hansen, Huang and Shek (2012) (Realized GARCH), Engle, Ledoit and Wolf (2019) (DCC-NL)
- Python Quantlets of this chapter: Quantlets/Ch_08
  - QML with sandwich standard errors, component GARCH, GARCH-MIDAS, realised kernels, jump tests, HAR/HARQ, Realized GARCH, DCC/cDCC and nonlinear shrinkage written out in `numpy`
- Lecture notebook: open in Google Colab

Data used in this chapter

Series	Source	Use
Daily realised measures of six equity indices (5-minute RV, BV, realised kernel, open-to-close return), 2000–2022	Heber, Lunde, Shephard and Sheppard (2009), v0.3	HAR, RGARCH
Bitcoin and Ether: one-minute prices 2018–2026, one-second prices August 2026	Binance public data (data.binance.vision); measures computed by us	noise, jumps, HARQ
S&P 500, DAX, BET, Bitcoin, 14 US stocks and 3 equity ETFs, daily	EODHD	QML, MIDAS, DCC
EUR/RON reference rate	BNR	QML
US industrial production, PPI finished goods (monthly)	FRED (INDPRO, WPSFD49207)	GARCH-MIDAS

- The Oxford-Man library ended in February 2022 (archived copy, cited as its terms require); Binance one-minute data extend the realised measures to 18 September 2026; a calibrated simulation gives the theory a known truth
- RV: realised variance; BV: bipower variation; PPI: producer price index

From the trading floor to the tick



New York Stock Exchange, 1908

Photo: Helen D. Van Eaton, Leslie's Monthly Magazine (1908); public domain; Wikimedia Commons

- 1982–1986: ARCH and GARCH Engle (1982), Bollerslev (1986): volatility as a filtered conditional variance of daily returns
- 1992–2004: QML theory Bollerslev and Wooldridge (1992), Lee and Hansen (1994), Lumsdaine (1996), Francq and Zakoïan (2004); components Engle and Lee (1999); DCC Engle (2002)
- 1998–2008: realised variance Andersen and Bollerslev (1998), Andersen, Bollerslev, Diebold and Labys (2003), Barndorff-Nielsen and Shephard (2002); bipower variation Barndorff-Nielsen and Shephard (2004); noise and kernels Zhang, Mykland and Aït-Sahalia (2005), Barndorff-Nielsen, Hansen, Lunde and Shephard (2008)
- 2009–2019: HAR Corsi (2009), Realized GARCH Hansen, Huang and Shek (2012), GARCH-MIDAS Engle, Ghysels and Sohn (2013), HARQ Bollerslev, Patton and Quaedvlieg (2016), DCC-NL Engle, Ledoit and Wolf (2019)
- Today: high-frequency data are cheap; the questions are statistical (noise, jumps, dimension, evaluation)

From TSA to this chapter

- Known (TSA, Chapter 5): GARCH(1,1), GJR, EGARCH, the news impact curve, ML with Gaussian and Student-t errors, QLIKE; (TSA, Chapter 14): CCC and DCC
- New here
  - why Gaussian ML still works when returns are not Gaussian, and which standard errors are then valid
  - volatility with a slow, macro-driven component; volatility measured from intraday prices, with its own sampling error
  - models that use realised measures; evaluation against a noisy target; covariance matrices with many assets
- Stochastic volatility (a latent log-variance with its own shock): Chapter 6; Markov-switching GARCH: Chapter 7; VaR and ES backtesting: Chapter 9; rough volatility: Chapter 10

The Gaussian quasi-likelihood (1/2): the model

- The return shock is a volatility times a standardised shock; the variance follows a GARCH(1,1) recursion

$$\varepsilon_t = \sigma_t(\theta_0)\,\eta_t, \qquad \sigma_t^2(\theta) = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2(\theta)$$

- Notation
  - ε_t: the demeaned return of day t; $\sigma_t^2(\theta)$: its variance conditional on the past $\mathcal{F}_{t-1}$ (the information up to day $t-1$)
  - η_t: i.i.d. standardised shock, $\mathbb{E}\eta_t = 0$, $\mathbb{E}\eta_t^2 = 1$; its law (Normal, Student-t, skewed) is unknown
  - $\theta = (\omega, \alpha, \beta)$, true value θ_0: $\omega > 0$ sets the level, $\alpha \geq 0$ the reaction to yesterday's squared shock, $\beta \geq 0$ the persistence of yesterday's variance
- The recursion starts from an arbitrary σ_1^2; the effect of the start vanishes geometrically when $\beta < 1$

The Gaussian quasi-likelihood (2/2): estimator and score

- QMLE: maximise the Gaussian log-likelihood, used as an estimating equation even if η_t is not Gaussian

$$\hat{\theta} = \arg\max_{\theta} \sum_{t=1}^{T} \ell_t(\theta), \qquad \ell_t(\theta) = -\frac{1}{2}\Big(\ln \sigma_t^2(\theta) + \frac{\varepsilon_t^2}{\sigma_t^2(\theta)} \Big)$$

  - ℓ_t: the Gaussian log-density of day t without its constant; T: number of days; $\hat{\theta}$: the estimate
- The score, the gradient of ℓ_t, has conditional mean zero at the true value

$$s_t(\theta) = \frac{\partial \ell_t}{\partial \theta} = \frac{1}{2}\Big(\frac{\varepsilon_t^2}{\sigma_t^2} - 1 \Big) \frac{1}{\sigma_t^2} \frac{\partial \sigma_t^2}{\partial \theta}$$

  - at θ_0, $\varepsilon_t^2/\sigma_t^2 = \eta_t^2$, hence $\mathbb{E}(s_t|\mathcal{F}_{t-1}) = 0$ because $\mathbb{E}\eta_t^2 = 1$
  - only the conditional variance must be right; the shape of the law of η_t is never used
- The score is a martingale difference (conditional mean zero given the past)
  - a CLT for martingales gives asymptotic normality without independence of ε_t

Consistency and asymptotic normality (1/2): stationarity and consistency

- GARCH(1,1) is strictly stationary if and only if the log of the daily variance multiplier has a negative mean

$$\mathbb{E}\ln(\alpha_0\eta_t^2+\beta_0)<0$$

  - $\alpha_0\eta_t^2+\beta_0$: the factor by which a shock to the variance is carried from one day to the next; α_0,β_0: true values
  - by Jensen's inequality, weaker than $\alpha_0+\beta_0<1$ (finite variance): IGARCH ($\alpha_0+\beta_0=1$) is strictly stationary, and $\mathbb{E}\varepsilon_t^2$ may be infinite
  - Lee and Hansen (1994), Lumsdaine (1996): GARCH(1,1), including IGARCH; Berkes, Horváth and Kokoszka (2003), Francq and Zakoïan (2004): GARCH(p,q) under strict stationarity
- Consistency of $\hat{\theta}$ Francq and Zakoïan (2004)
  - conditions: strict stationarity, θ_0 in a compact parameter set, identifiability (η_t^2 is not a constant)
  - no moment of ε_t is needed
- Even explosive GARCH ($\mathbb{E}\ln(\alpha_0\eta^2+\beta_0)>0$): $(\hat{\alpha},\hat{\beta})$ remain consistent and asymptotically normal, ω is not identified Jensen and Rahbek (2004)

Consistency and asymptotic normality (2/2): the limit law

- With θ_0 interior to the parameter set and a finite fourth moment of η_t, the estimator is asymptotically normal

$$\sqrt{T}(\hat{\theta} - \theta_0) \to N\big(0, (\kappa_\eta - 1) J^{-1}\big), \qquad J = \mathbb{E}\Big[\frac{1}{\sigma_t^4} \frac{\partial \sigma_t^2}{\partial \theta} \frac{\partial \sigma_t^2}{\partial \theta'}\Big]$$

  - $\kappa_\eta = \mathbb{E}\eta_t^4$: kurtosis of the shock (3 for the Normal distribution); $\kappa_\eta - 1 = \mathrm{Var}(\eta_t^2)$
  - J: a 3×3 matrix, the mean outer product of the relative sensitivities $\sigma_t^{-2} \partial \sigma_t^2 / \partial \theta$; $\to$: convergence in distribution as $T \to \infty$
  - reading: the sampling variance of $\hat{\theta}$ grows linearly with the kurtosis of the shocks
- Boundary: if $\alpha_0 = 0$, θ_0 is not interior and the limit is the projection of a normal vector on a cone
  - this is why TSA, Chapter 5 tests ARCH effects one-sided

The sandwich and the Bollerslev–Wooldridge standard errors (1/2)

- General QML result: the asymptotic variance is a "sandwich" of two matrices

$$\sqrt{T}(\hat{\theta} - \theta_0) \to N(0, A^{-1} B A^{-1}), \qquad A = -\mathbb{E}\frac{\partial^2 \ell_t}{\partial\theta\,\partial\theta'}, \qquad B = \mathbb{E}\, s_t s_t'$$

  - A: the expected negative Hessian (curvature of the log-likelihood); B: the variance of the score s_t
  - if the likelihood is the true one, $A = B$ (information equality) and the variance reduces to A^{-1}; under QML, $A \neq B$

- For GARCH both matrices are proportional to J (▸ Appendix)

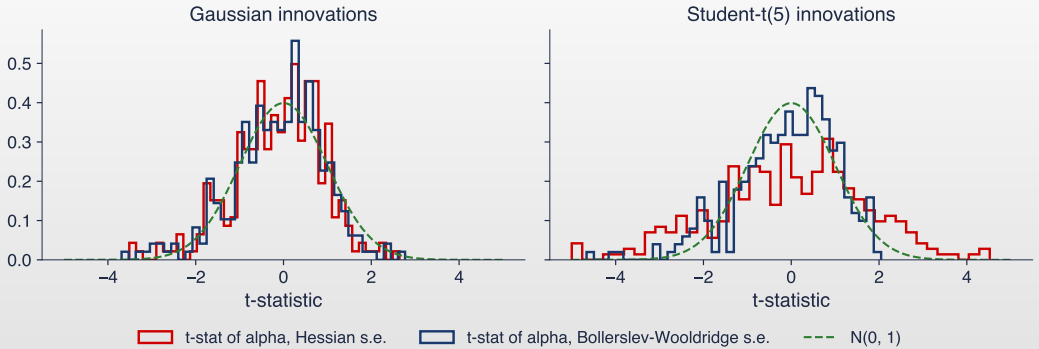
$$A = \tfrac{1}{2} J, \qquad B = \tfrac{\kappa_\eta - 1}{4} J, \qquad A^{-1} B A^{-1} = (\kappa_\eta - 1) J^{-1}$$

  - the sandwich recovers the limit law of the previous slide, for any law of η_t with $\kappa_\eta < \infty$

The sandwich and the Bollerslev–Wooldridge standard errors (2/2)

- Three ways to compute standard errors (s.e.)
  - Hessian only: $A^{-1} = 2J^{-1}$, correct only if $\kappa_\eta = 3$; too small by the factor $\sqrt{(\kappa_\eta - 1)/2}$ when $\kappa_\eta > 3$
  - outer product of the scores (OPG): $B^{-1} = \frac{4}{\kappa_\eta - 1} J^{-1}$, even smaller when $\kappa_\eta > 3$
  - sandwich Bollerslev and Wooldridge (1992): $\hat{A}^{-1}\hat{B}\hat{A}^{-1}$, valid whatever the law of η_t (with $\kappa_\eta < \infty$)
- Estimates used by Bollerslev and Wooldridge
  - $\hat{A}$: the observed Hessian of $-\frac{1}{T}\sum_t \ell_t$ at $\hat{\theta}$; $\hat{B} = \frac{1}{T}\sum_t \hat{s}_t \hat{s}_t'$, with $\hat{s}_t = s_t(\hat{\theta})$ the estimated score
- Robust Wald and score tests use the sandwich
  - the quasi-likelihood-ratio (quasi-LR) statistic is no longer χ^2, but a weighted sum of χ^2_1 variables (chi-square with one degree of freedom)

Hessian against sandwich: a Monte Carlo



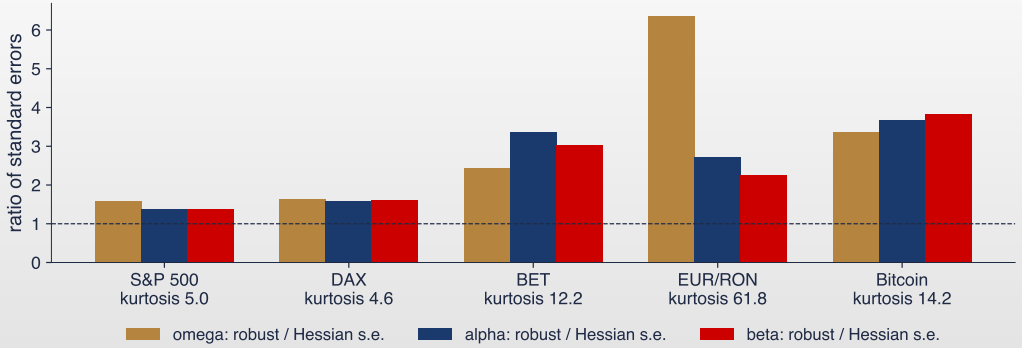
- GARCH(1,1) with $(\omega, \alpha, \beta) = (0.05, 0.08, 0.90)$, $T = 2\,000$, 300 replications; Gaussian QML in each; histogram of $(\hat{\alpha} - \alpha_0)/\widehat{\text{se}}$

ATS_ch8_qmle

Interpreting the Monte Carlo

- Gaussian η: coverage of nominal 95% intervals for α is 95.0% (Hessian) and 94.3% (sandwich): both fine
- Student-$t(5)$, $\kappa_\eta = 9$: Hessian intervals cover 73.0% for α and 74.7% for β; sandwich intervals 91.7% and 92.3%
- The theory predicts the factor $\sqrt{(9-1)/2} = 2$ between the two standard errors; the point estimates are the same
- With fat tails, a Hessian-based t-test of α rejects far too often: report sandwich standard errors by default

Robust and naive standard errors in five markets



⊡ Gaussian QML of GARCH(1,1), daily returns 2010–2026; bars: sandwich s.e. divided by Hessian s.e.; labels: kurtosis $\hat{\kappa}_\eta$ of the standardised residuals

ATS_ch8_qmle

Interpreting the five markets

- S&P 500: $\hat{\alpha} = 0.154$ with s.e. 0.013 (Hessian) against 0.018 (sandwich); $\hat{\kappa}_\eta = 5.0$, theoretical factor 1.42
- BET: $\hat{\kappa}_\eta = 12.2$, the sandwich s.e. of $\hat{\alpha}$ is 3.36 times larger; EUR/RON (managed float): 2.71 times, $\hat{\kappa}_\eta = 61.8$
- The ratios track $\sqrt{(\hat{\kappa}_\eta - 1)/2}$ only roughly: in the data η_t is not i.i.d., and the sandwich does not need it to be
- Student-t ML raises the log-likelihood by 124.4 (S&P 500) and 233.0 (BET), but is consistent only if the t shape is right; QML needs only the variance equation

GARCH with exogenous variables

- GARCH-X: yesterday's value of an observed variable enters the variance equation

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 + \pi' x_{t-1}$$

  - x_{t-1}: vector of exogenous variables known at the end of day $t-1$; π: their loadings ($\pi' x$: the weighted sum)
  - positivity of σ_t^2: $x_{t-1} \geq 0$ and $\pi \geq 0$, or a specification in logs
- Candidates for x_t: a realised measure (leads to HEAVY and Realized GARCH below), the VIX, a macro or policy-uncertainty index, trading volume
  - QML theory carries over if x_t is stationary and ergodic; its own dynamics need not be modelled for one-step forecasts
  - multi-step forecasts need a model for x_t: this is exactly what Realized GARCH and HEAVY add
- A slow variable (a monthly macro series) cannot enter a daily recursion at its own frequency: one needs a component structure (next slides)

The component GARCH of Engle and Lee (1/2): the model

- Engle and Lee (1999): the variance is a slowly moving long-run level q_t plus a transitory deviation from it

$$\sigma_t^2 = q_t + \alpha(\varepsilon_{t-1}^2 - q_{t-1}) + \beta(\sigma_{t-1}^2 - q_{t-1})$$

$$q_t = \omega + \rho q_{t-1} + \phi(\varepsilon_{t-1}^2 - \sigma_{t-1}^2)$$

  - q_t: long-run (permanent) component; ρ: its persistence, close to 1; ω: its intercept
  - $\sigma_t^2 - q_t$: transitory component; α, β: its reaction and persistence, with $\alpha + \beta < \rho$
  - $\varepsilon_{t-1}^2 - \sigma_{t-1}^2$: the variance surprise of yesterday (mean zero); ϕ: how much of it moves the long-run level

- Both components are driven by the same shock: the model is a restricted GARCH(2,2)

The component GARCH of Engle and Lee (2/2): two speeds

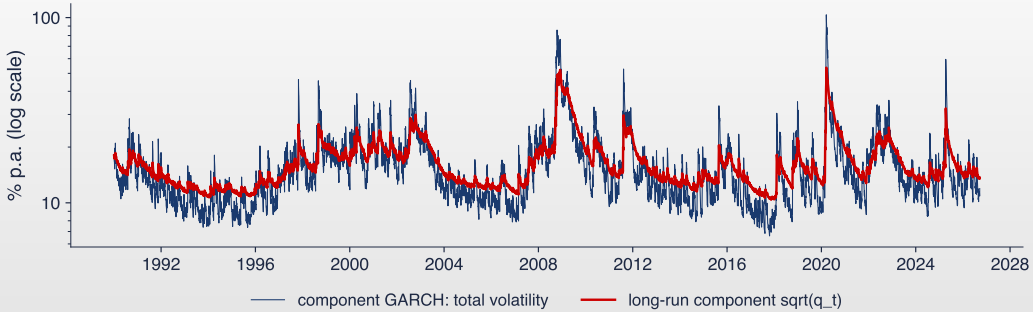
- Half-life: the number of days after which half of a shock to a component has died out

$$\mathrm{HL}_q = \frac{\ln(1/2)}{\ln \rho}, \qquad \mathrm{HL}_{\sigma-q} = \frac{\ln(1/2)}{\ln(\alpha+\beta)}$$

  - HL_q: half-life of the long-run component; $\mathrm{HL}_{\sigma-q}$: half-life of the transitory component
  - a persistence $\rho = 0.99$ gives about 69 days; $\alpha + \beta = 0.9$ gives about 6.6 days
  - GARCH(1,1) forces a single half-life, $\ln 0.5 / \ln(\alpha+\beta)$

- Two exponentials approximate a slowly decaying (hyperbolic) autocorrelation of squared returns over a finite range of lags
  - long memory proper: Chapter 10

Component GARCH for the S&P 500



- Daily returns since 1990, $T = 9\,240$; Gaussian QML; annualised total volatility $\sqrt{252\sigma_t^2}$ and long-run component $\sqrt{252q_t}$

ATS_ch8_components

Interpreting the component model

- $\hat{\rho} = 0.995$, $\hat{\phi} = 0.026$; transitory part $\hat{\alpha} + \hat{\beta} = 0.950$: half-lives 150 days (long run) and 13.6 days (short run)
- GARCH(1,1) on the same data: persistence 0.983, one half-life of 41 days, a compromise between the two
- Quasi-LR statistic against GARCH(1,1): 44.1; under the null $\phi = 0$ the parameter ρ is not identified (Davies problem), so the χ^2_2 reference is only indicative
- After 2008, 2020 and 2025 the long-run level stays high for months while daily volatility has already fallen: this matters for horizons beyond a few weeks

GARCH-MIDAS (1/3): a macro-driven long-run component



Eric Ghysels, 2019

Photo: Eghysels (2019); CC BY-SA 4.0; Wikimedia Commons

- Engle, Ghysels and Sohn (2013): the daily variance is the product of a monthly level τ_t and a daily factor $g_{i,t}$

$$r_{i,t} = \mu + \sqrt{\tau_t \, g_{i,t}} \, \eta_{i,t}$$

  - $r_{i,t}$: return of day i of month t; μ: mean return; $\eta_{i,t}$: i.i.d. shock with mean 0 and variance 1

- Short run: a GARCH(1,1) with mean 1, re-scaled by τ_t

$$g_{i,t} = (1 - \alpha - \beta) + \alpha \frac{(r_{i-\mathbf{1},t} - \mu)^{\mathbf{2}}}{\tau_t} + \beta g_{i-\mathbf{1},t}$$

  - α, β: reaction and persistence of the daily component, as in GARCH(1,1)

GARCH-MIDAS (2/3): the long-run component

- Long run: MIDAS (mixed-data sampling) regression of $\ln \tau_t$ on K lags of a monthly variable

$$\ln \tau_t = m + \theta \sum_{k=1}^{K} \varphi_k(w)\, X_{t-k}, \qquad \varphi_k(w) \propto (1 - k/K)^{w-1}$$

  - X_{t-k}: the monthly driver k months ago; m: intercept; θ: effect of the driver (sign: pro- or countercyclical)
  - $\varphi_k(w)$: beta lag weights, normalised to sum to 1 and restricted to decay Ghysels, Santa-Clara and Valkanov (2006); $w \geq 1$: the larger w, the faster the decay; $w = 1$: equal weights
- Choices of X
  - realised variance of past months, in levels: $\tau_t = m + \theta \sum_k \varphi_k(w) \mathrm{RV}_{t-k}$, with RV_{t-k} the sum of squared daily returns of month $t - k$
  - macro data: industrial production growth, PPI inflation

GARCH-MIDAS (3/3): identification, estimation and the variance ratio

- τ_t is constant within the month and $\mathbb{E} g_{i,t} = 1$: the split is identified by the different frequencies, without a filter for a latent variable
  - Gaussian QML of all parameters $(\mu, \alpha, \beta, m, \theta, w)$ in one step; sandwich standard errors
- Variance ratio: the share of the variation of log volatility explained by the long-run component

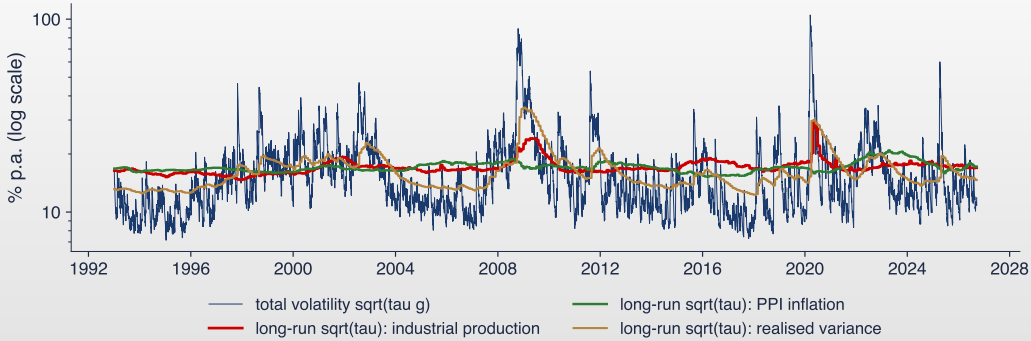
$$\mathrm{VR} = \frac{\mathrm{Var}(\ln \tau_t)}{\mathrm{Var}(\ln(\tau_t \, g_{i,t}))} \in [0, 1]$$

  - VR close to 0: the long-run component is almost flat; close to 1: it carries most of the variation
  - a macro variable can be significant ($\hat{\theta} \neq 0$) and still explain little of daily volatility (small VR)
- Inference on w is nonstandard when $\theta = 0$ (w is then not identified) and at the boundary $w = 1$
- Extensions: two-sided weights, several regressors, a second (daily) component Conrad and Kleen (2020); forecasts beyond one month use only the long-run part

Case study: Engle, Ghysels and Sohn (2013) on our data

- **Question** of Engle, Ghysels and Sohn (2013): does the stock market's long-run volatility move with macroeconomic fundamentals (output growth, inflation)?
  - their sample of US stock returns is much longer than ours; their benchmark long-run component is a fixed-window realised variance
- **Our replication**: S&P 500 daily returns, common sample from 04.01.1993, $T = 8\,482$; $K = 36$ monthly lags; restricted beta weights
  - three long-run drivers: monthly realised variance (sum of squared daily returns), industrial production growth (FRED INDPRO), PPI inflation (FRED WPSFD49207)
  - all models against GARCH(1,1) on the same days; QML with sandwich standard errors
- Extension for a project: Romanian industrial production and the BET (Seminar 8, B5); real-time vintages of the macro data

GARCH-MIDAS for the S&P 500



- Annualised $\sqrt{252\tau_t g_{i,t}}$ of the industrial-production model and the long-run components $\sqrt{252\tau_t}$ of the three models

ATS_ch8_components

GARCH-MIDAS estimates

Long-run driver	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\theta}$ (s.e.)	$\hat{w}$	VR (%)	BIC
realised variance (levels)	0.119	0.841	0.0225 (0.0058)	5.939	25.1	22710.0
industrial production growth	0.110	0.870	−0.551 (0.353)	3.692	4.5	22732.5
PPI inflation	0.108	0.873	0.753 (1.260)	1.000	2.2	22733.3
GARCH(1,1)						22721.9

- Same days for all models; BIC $= -2 \ln L + k \ln T$ (lower is better); s.e.: Bollerslev–Wooldridge

Interpreting GARCH-MIDAS

- Realised variance as the slow driver: $\hat{\theta} = 0.0225$ ($t = 3.86$), VR = 25.1%, BIC below GARCH(1,1): a long-run component exists
- Industrial production: $\hat{\theta} = -0.551$ ($t = -1.56$), the countercyclical sign of Engle, Ghysels and Sohn (2013), but VR only 4.5% and a higher BIC than GARCH(1,1)
- PPI inflation: $\hat{w}$ at the boundary 1 (flat weights) and $\hat{\theta}$ imprecise: no evidence in 1993–2026
- Replication verdict: the qualitative sign survives, the strength does not; macro effects need long samples with deep recessions, which is why the original study uses a much longer history

Prices as Itô semimartingales (1/2): the model

- Within day t, the log price moves by a drift, a Brownian part scaled by the spot volatility, and jumps

$$dX_s = \mu_s\, ds + \sigma_s\, dW_s + dJ_s, \qquad s \in [0,1]$$

  - X_s: log price at intraday time s (the day is rescaled to $[0,1]$); μ_s: drift; W_s: standard Brownian motion
  - σ_s: stochastic spot (instantaneous) volatility; J_s: a jump process with finitely many jumps per day, ΔJ_s: the jump at time s

- Quadratic variation of the day: the diffusive part plus the squared jumps

$$\mathrm{QV}_t = \mathrm{IV}_t + \sum_{s \le 1} (\Delta J_s)^2, \qquad \mathrm{IV}_t = \int_0^1 \sigma_s^2\, ds$$

  - IV_t: integrated variance, the average of the spot variance over the day
  - without jumps and with σ independent of W: $r_t | \mathrm{IV}_t \sim N(\int_0^1 \mu_s\, ds, \mathrm{IV}_t)$, so IV is the variance that matters for the daily return r_t

Prices as Itô semimartingales (2/2): realised variance

- Realised variance: the sum of the n squared intraday returns of the day

$$\mathrm{RV}_t = \sum_{i=1}^{n} r_{t,i}^2, \qquad r_{t,i} = X_{i/n} - X_{(i-1)/n}$$

  - n: number of intraday intervals (78 five-minute returns in a 6.5-hour session); $r_{t,i}$: log return over the i-th interval
  - $\mathrm{RV}_t \to \mathrm{QV}_t$ in probability as $n \to \infty$: RV measures the total (diffusive plus jump) variation
- The drift does not matter in the limit: it is $O(1/n)$ per interval, while the Brownian part is $O(n^{-1/2})$
  - volatility becomes observable *ex post*, without a model Andersen, Bollerslev, Diebold and Labys (2003), Barndorff-Nielsen and Shephard (2002)

The central limit theorem for realised variance (1/2)

- Barndorff-Nielsen and Shephard (2002): without jumps, the error of RV shrinks at rate $\sqrt{n}$ and is mixed normal

$$\sqrt{n}\,(\mathrm{RV}_t - \mathrm{IV}_t) \to MN(0, 2\,\mathrm{IQ}_t), \qquad \mathrm{IQ}_t = \int_0^1 \sigma_s^4\, ds$$

  - IQ_t: integrated quarticity, the daily average of σ_s^4; it sets the size of the error
  - *MN* (mixed normal): normal given the path of σ, with a random variance $2\,\mathrm{IQ}_t$
  - the convergence is *stable* in law: one may divide by a random, consistently estimated $\sqrt{\mathrm{IQ}_t}$ and keep the limit

- Rate $\sqrt{n}$: with $n = 78$ and volatility constant within the day, the relative error is about $\sqrt{2/78} \approx 16\%$
  - the measurement error $\mathrm{RV}_t - \mathrm{IV}_t$ is heteroskedastic (it scales with IQ_t): HARQ, below, is built on this fact

The central limit theorem for realised variance (2/2): feasible intervals

- Feasible version: estimate IQ by the realised quarticity RQ, then studentise

$$\widehat{\mathrm{IQ}}_t = \mathrm{RQ}_t = \frac{n}{3}\sum_{i=1}^{n} r_{t,i}^4, \qquad \frac{\mathrm{RV}_t - \mathrm{IV}_t}{\sqrt{\frac{2}{3}\sum_i r_{t,i}^4}} \to N(0,1)$$

  - the factor $\frac{n}{3}$ makes RQ unbiased under constant volatility, because $\mathbb{E}Z^4 = 3$ for $Z \sim N(0,1)$
  - 95% interval for IV_t: $\mathrm{RV}_t \pm 1.96\sqrt{\frac{2}{3}\sum_i r_{t,i}^4}$; it may contain negative values

- Log version (delta method): an interval for $\ln \mathrm{IV}_t$

$$\ln \mathrm{RV}_t \pm 1.96\,\frac{\sqrt{\frac{2}{3}\sum_i r_{t,i}^4}}{\mathrm{RV}_t}$$

  - exponentiated, it never contains negative values and has better coverage in finite samples

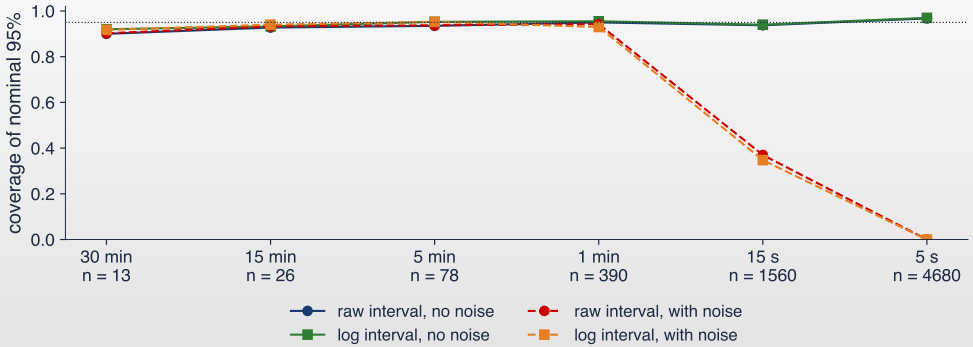
Simulated prices with known integrated variance

- One-second grid, $n = 23\,400$ seconds a day; the log spot variance is the sum of two Gaussian AR(1) factors

$$\ln \sigma_s^2 = c + f_s^{(1)} + f_s^{(2)}$$

  - c: constant fixing the mean; $f^{(1)}$: slow factor (half-life 60 days, stationary s.d. 0.75); $f^{(2)}$: fast factor (half-life 2 days, s.d. 0.45)
  - mean IV = 1 (%^{2} a day), the mean of the S&P 500 realised kernel in 2000–2022; leverage: correlation -0.6 between price shocks and shocks to the fast factor
- Jumps: Poisson with 0.08 jumps a day, sizes $N(0, 0.8^2)$ (%): jumps make about 5.6% of quadratic variation
- Noise: we observe $Y_s = X_s + u_s$, with u_s i.i.d. $N(0, \omega^2)$ and $\omega = 0.004\%$
  - Y_s: observed log price; X_s: efficient log price; u_s: microstructure noise with standard deviation ω; at one second, $2n\omega^2 = 0.75$ (slide Microstructure noise)
- Each estimator below is compared with the true IV or QV of the same simulated day

Coverage of the feasible CLT



- 400 simulated days without jumps; 95% intervals for IV_t (raw and log) from returns sampled every 30 minutes to every 5 seconds, without and with noise

ATS_ch8_realised_measures

Interpreting the CLT coverage

- Without noise, coverage approaches 95% as n grows: raw interval 90.0% at $n = 13$ and 95.0% at $n = 390$; the log interval is closer at small n (92.0%)
- With noise, coverage collapses once sampling is too fine: 37.0% at 15 seconds and 0.0% at 5 seconds; at 1 minute it is still 94.2%
- The interval shrinks like $n^{-1/2}$ while the noise bias grows like n: the CLT is a statement about the efficient price, not about the observed one
- Practical rule: 5-minute returns for liquid assets unless a noise-robust estimator is used

Where the realised measures come from

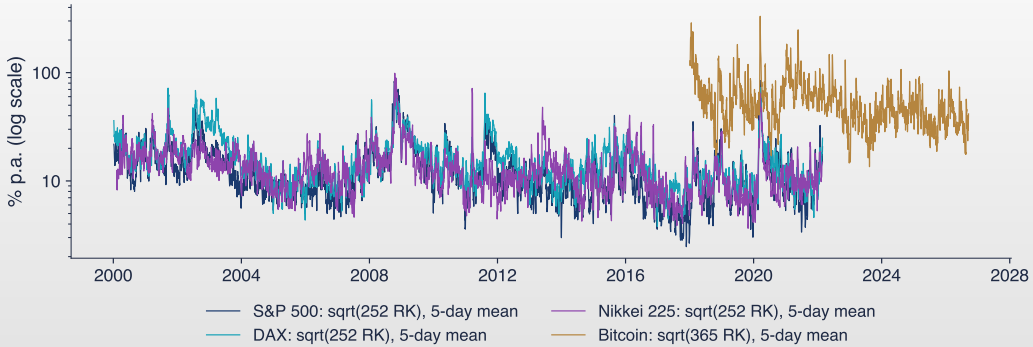


Ole E. Barndorff-Nielsen, 2007

Photo: Thomas Steiner (2007); CC BY-SA 2.5; Wikimedia Commons

- Heber, Lunde, Shephard and Sheppard (2009), version 0.3: daily RV, BV, realised kernels and open-to-close returns for 31 indices, 2000 to 25 February 2022, built from cleaned tick data
- its terms: free use if the library and its version are cited; the archived copy keeps the 2022 file available after the shutdown
- Bitcoin and Ether trade 24/7 and Binance publishes all one-minute and one-second prices: we compute RV, BV, RQ, tripower quarticity and a realised kernel for each UTC day, 2018–2026
- No free source covers equity intraday data after 2022: a project can rebuild them from a licensed feed

Realised volatility across markets



⊡ Annualised realised kernel, 5-day means: S&P 500, DAX and Nikkei 225 (Oxford-Man, 2000–2022); Bitcoin (Binance one-minute prices, 2018–2026, 365 days a year)

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Interpreting the realised volatilities

- Average annualised volatility from the kernel: S&P 500 16.2%, DAX 19.5%, Nikkei 16.6%, Bitcoin 68.4%
- The equity series move together (2002, 2008, 2011, 2020): a common long-run factor, the multivariate theme of the last section
- Open-to-close measures miss the overnight return: for daily risk one adds the squared overnight return or scales the measure
- Bitcoin has no overnight gap, but its daily volatility is three to four times higher and falls after 2023

Microstructure noise (1/2): the bias of RV

- The observed log price is the efficient price plus a noise term

$$Y_{i/n} = X_{i/n} + u_i$$

  - u_i: i.i.d. noise with variance ω^2, independent of X; sources: bid–ask bounce, price discreteness, stale quotes

- RV computed from the observed prices at n intervals, $\mathrm{RV}_t^{(n)}$, is biased upwards, and the bias grows with n

$$\mathbb{E}(\mathrm{RV}_t^{(n)}|X) = \mathrm{IV}_t + 2n\omega^2, \qquad \mathrm{Var}(\mathrm{RV}_t^{(n)}|X) \approx 4n\,\mathbb{E}u^4$$

  - each observed return contains $u_i - u_{i-1}$, with variance $2\omega^2$; summed over n returns this gives $2n\omega^2$
  - RV diverges as $n \to \infty$: sampling as often as possible is not optimal
  - derivation of the bias and of the variance: Appendix

Microstructure noise (2/2): diagnostics

- Two consequences of i.i.d. noise
  - $\hat{\omega}^2 = \mathrm{RV}_t^{(n)}/(2n)$ at the highest frequency estimates the noise variance, since $2n\omega^2$ dominates IV there
  - returns of the observed price have first-order autocovariance $-\omega^2$: negative autocorrelation is the signature of i.i.d. noise
- Signature plot: average $\mathrm{RV}^{(n)}$ against the sampling interval
  - a flat region shows the intervals where noise no longer matters
- Real noise is not i.i.d.: it is autocorrelated and correlated with the efficient price, especially in quote data Hansen and Lunde (2006b)

Optimal sparse sampling

- Mean square error (MSE) of $\mathrm{RV}^{(n)}$ as an estimator of IV: sampling variance falls, noise bias rises with n

$$\mathrm{MSE}(n) \approx \underbrace{\frac{2\,\mathrm{IQ}}{n}}_{\text{variance}} + \underbrace{(2n\omega^2)^2}_{\text{bias}^2} + 4n\,\mathbb{E}u^4 + \dots$$

- Minimising the first two terms gives the optimal number of intervals Bandi and Russell (2008)

$$n^* \approx \left(\frac{\mathrm{IQ}}{4\omega^4}\right)^{1/3}$$

  - the optimal grid is coarser when noise (ω^2) is large relative to volatility (IQ)
  - example (Seminar 8, A3): IV = IQ = 1, $\omega^2 = 1.6 \times 10^{-5}$ gives $n^* \approx 992$, a return every 23.6 seconds

- The best sparse RV converges only at rate $n^{1/6}$ and discards most data: this motivates estimators that use all observations
- Subsampling: average the RV of the K offset grids (starting at second $0, 1, \dots, K-1$); same bias, smaller variance

Two scales and realised kernels (1/2)

- Two-scales RV Zhang, Mykland and Aït-Sahalia (2005): the fast scale estimates the noise bias of the slow one and removes it

$$\mathrm{TSRV} = \overline{\mathrm{RV}}^{(K)} - \frac{\bar{n}}{n}\,\mathrm{RV}^{(n)}, \qquad \bar{n} = \frac{n - K + 1}{K}$$

  - $\overline{\mathrm{RV}}^{(K)}$: subsampled RV on K offset sparse grids (slow scale), each with about $\bar{n}$ returns; $\mathrm{RV}^{(n)}$: RV on all n returns (fast scale)
  - the bias of the slow scale is $2\bar{n}\omega^2 = \frac{\bar{n}}{n} \cdot 2n\omega^2$: the correction subtracts it; rate $n^{1/6}$
- Pre-averaging Jacod et al. (2009): average returns over blocks before squaring
  - rate $n^{1/4}$, the optimal one in the presence of noise; in practice similar to the realised kernel

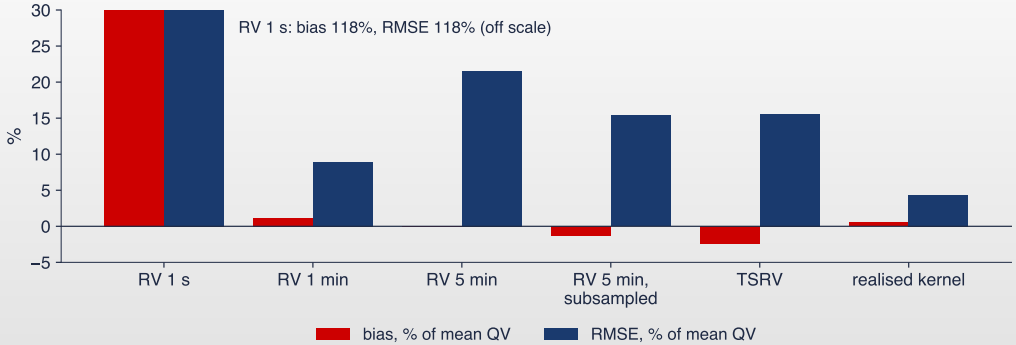
Two scales and realised kernels (2/2): the realised kernel

- Realised kernel Barndorff-Nielsen, Hansen, Lunde and Shephard (2008): RV plus weighted realised autocovariances

$$\mathrm{RK}_t = \gamma_0 + \sum_{h=1}^{H} k\Big(\frac{h}{H+1}\Big)(\gamma_h + \gamma_{-h}), \qquad \gamma_h = \sum_i r_{t,i}\, r_{t,i-h}$$

  - γ_h: realised autocovariance at lag h ($\gamma_0 = \mathrm{RV}$); H: bandwidth, the number of lags used; $k(\cdot)$: weight function with $k(0) = 1$, $k(1) = 0$
  - Parzen weights: $k(x) = 1 - 6x^2 + 6x^3$ for $x \le \frac{1}{2}$ and $2(1-x)^3$ for $\frac{1}{2} < x \le 1$
  - the negative γ_1 created by noise cancels the bias $2n\omega^2$: the same idea as HAC variance estimation (Chapter 0)
- Properties
  - nonnegative by construction; rate $n^{1/5}$ with $H = c^*\xi^{4/5}n^{3/5}$, where $\xi^2 = \omega^2/\sqrt{\mathrm{IQ}}$ is the noise-to-signal ratio and $c^* = 3.51$ for Parzen Barndorff-Nielsen et al. (2009)
  - it also corrects serially dependent noise of short range, of either sign

Estimators against the true quadratic variation



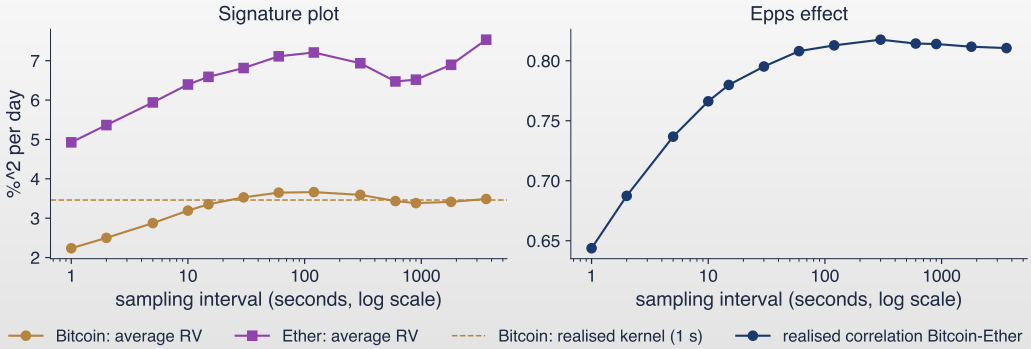
- 300 simulated days with jumps and noise; bias and root mean square error relative to the mean QV; TSRV with $K = 300$ seconds; RK with the BNHLS bandwidth (median $H = 31$)

ATS_ch8_realised_measures

Interpreting the estimator comparison

- RV at one second is dominated by noise: bias 117.8% of QV; at one minute the bias is 1.2% and the RMSE 8.9%
- Five-minute RV is unbiased but noisy (RMSE 21.6%); subsampling and TSRV reduce it to 15.4% and 15.6%
- The realised kernel uses all 23 400 returns: bias 0.5%, RMSE 4.4%, the best of the six
- All estimators here target QV, jumps included; separating the jump part needs the next section

Signature plot and Epps effect: Bitcoin and Ether



- Binance one-second prices, August 2026 (31 days); average daily RV by sampling interval (subsampled); realised correlation of the two coins by interval

ATS_ch8_realised_measures

Interpreting the signature plot

- Bitcoin: RV is 2.24 at one second, 3.65 at one minute, 3.59 at five minutes: the signature *rises*, the opposite of the i.i.d.-noise prediction
- Reason: 52% of one-second returns are zero (no trade or the same price) and prices adjust gradually, so one-second returns are positively autocorrelated
- The realised kernel on one-second data (3.46, median $H = 22$) adds the positive autocovariances back and lands on the flat region of 1–30 minutes
- Epps effect Epps (1979): realised correlation 0.64 at one second against 0.82 at five minutes; asynchronous trading biases covariances towards zero at high frequency

Bipower variation (1/2)

- Barndorff-Nielsen and Shephard (2004): sum of products of adjacent absolute returns, rescaled

$$\mathrm{BV}_t = \mu_1^{-2} \frac{n}{n-1} \sum_{i=2}^{n} |r_{t,i}|\,|r_{t,i-1}|, \qquad \mu_1 = \mathbb{E}|Z| = \sqrt{2/\pi}$$

  - $Z \sim N(0,1)$; μ_1^{-2} makes BV_t consistent for IV_t; $\frac{n}{n-1}$ corrects for the $n-1$ products
  - $\mathrm{BV}_t \to \mathrm{IV}_t$ even with (finite-activity) jumps: a jump enters only two products, each multiplied by a return of order $n^{-1/2}$
- Hence $\mathrm{RV}_t - \mathrm{BV}_t \to \sum_{s \le 1} (\Delta J_s)^2$: a nonparametric estimate of the jump variation of the day

Bipower variation (2/2): related estimators

- Jump-robust quarticity: the tripower quarticity

$$\mathrm{TQ}_t = n\,\mu_{4/3}^{-3}\,\frac{n}{n-2}\sum_{i=3}^{n}\prod_{j=0}^{2}|r_{t,i-j}|^{4/3}, \qquad \mu_{4/3} = \mathbb{E}|Z|^{4/3}$$

  - products of three adjacent absolute returns, each to the power 4/3; $\mathrm{TQ}_t \to \mathrm{IQ}_t$ even with jumps (RQ does not)
- Nearest-neighbour truncation (MedRV, MinRV) Andersen, Dobrev and Schaumburg (2012): smaller finite-sample bias from a jump and from zero returns
- Threshold (truncated) RV: drop the returns above $c\,n^{-\varpi}$
  - $c > 0$: a constant (a multiple of the local volatility); $\varpi \in (0, \frac{1}{2})$: the threshold shrinks more slowly than a diffusive return, so only jumps are removed

Testing for jumps (1/2): the ratio statistic

- Barndorff-Nielsen and Shephard (2006): without jumps, the difference RV − BV is also mixed normal

$$\sqrt{n}(\mathrm{RV}_t - \mathrm{BV}_t) \to MN(0, \vartheta\, \mathrm{IQ}_t), \qquad \vartheta = \mu_1^{-4} + 2\mu_1^{-2} - 5 = \frac{\pi^2}{4} + \pi - 5 \approx 0.609$$

  - ϑ: a constant that measures how much less efficient BV is than RV
- Ratio statistic Huang and Tauchen (2005): the relative jump contribution, studentised

$$z_t = \frac{(\mathrm{RV}_t - \mathrm{BV}_t)/\mathrm{RV}_t}{\sqrt{\frac{\vartheta}{n} \max\left(1, \mathrm{TQ}_t/\mathrm{BV}_t^2\right)}} \to N(0,1)$$

  - one-sided test: reject "no jump on day t" for large z_t
  - the ratio form and the max adjustment give the best size in their simulations; $\mathrm{TQ}_t/\mathrm{BV}_t^2 \geq 1$ by Jensen when volatility varies within the day

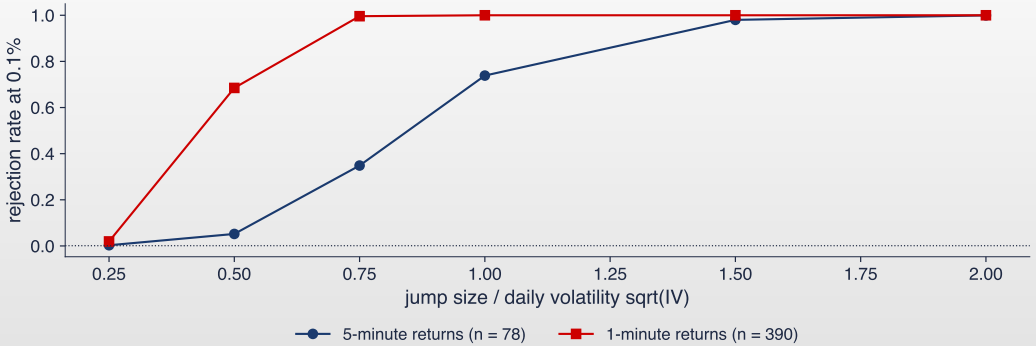
Testing for jumps (2/2): in practice

- A daily test answers "was there a jump today?"; locating it within the day needs a test on each return standardised by local BV Lee and Mykland (2008)
- Multiple testing: at level α over T days one expects αT false jump days
  - use a small α (0.1%) or a family-wise (FWER) or false-discovery-rate (FDR) correction (Chapter 1)
- Separating the continuous and jump parts of RV Andersen, Bollerslev and Diebold (2007)

$$J_t = \mathbb{1}(z_t > z_{1-\alpha})\,(\mathrm{RV}_t - \mathrm{BV}_t), \qquad C_t = \mathrm{RV}_t - J_t$$

  - $\mathbb{1}(\cdot)$: indicator, 1 if the test rejects and 0 otherwise; $z_{1-\alpha}$: the $1-\alpha$ quantile of $N(0,1)$
  - J_t: jump part, nonzero only on significant days; C_t: continuous part

Size and power of the ratio test



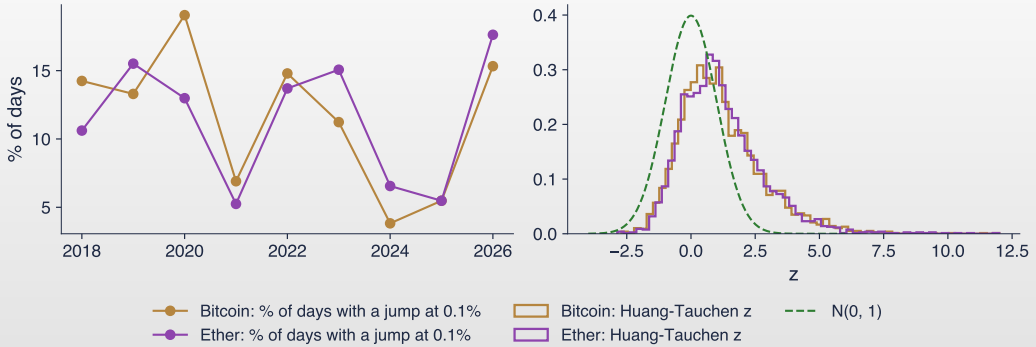
- 2 000 simulated days (two-factor SV, no jumps), then one jump of size $c\sqrt{\mathrm{IV}_t}$ at a random time; one-sided test at 0.1% with 5-minute and 1-minute returns

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Interpreting the size and power

- Size without jumps: 0.0% (5 minutes) and 0.2% (1 minute) against the nominal 0.1%; with i.i.d. noise of the calibrated size still 0.1% at 1 minute
- Power for a jump of half a daily standard deviation: 5% with 5-minute returns, 68% with 1-minute returns; for one standard deviation 74% and 100%
- The power depends on $c\sqrt{n}$: small jumps are absorbed into the diffusion at coarse sampling (Seminar 8, A6 gives the detection threshold)
- The trade-off with noise is the same as for RV: finer sampling raises power until noise and zero returns distort the statistic

Jump days of Bitcoin and Ether



- Huang–Tauchen test with 5-minute returns ($n = 288$ a UTC day), 2018–2026, $T = 3\,165$ days; left: share of jump days at 0.1% by year; right: distribution of z_t against $N(0, 1)$

ATS_ch8_jumps

Interpreting the crypto jump tests

- Bitcoin: 362 jump days (11.4%) where about 3.2 false rejections are expected; Ether: 355 days (11.2%)
- Yet the jumps are small: they make 3.0% (Bitcoin) and 2.3% (Ether) of total variation; the rest is continuous
- The whole distribution of z_t is shifted (mean 1.16, not 0): besides jumps, 24-hour intraday seasonality, zero returns and flash moves violate the assumptions of the null
- Before calling a day a jump day: standardise returns by an intraday volatility pattern, check zero returns, and control the number of tests

The HAR model of Corsi (1/2): the regression

- Corsi (2009): tomorrow's RV is a linear function of the daily, weekly and monthly averages of past RV

$$\mathrm{RV}_{t+1} = \beta_0 + \beta_d \mathrm{RV}_t + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + u_{t+1}$$

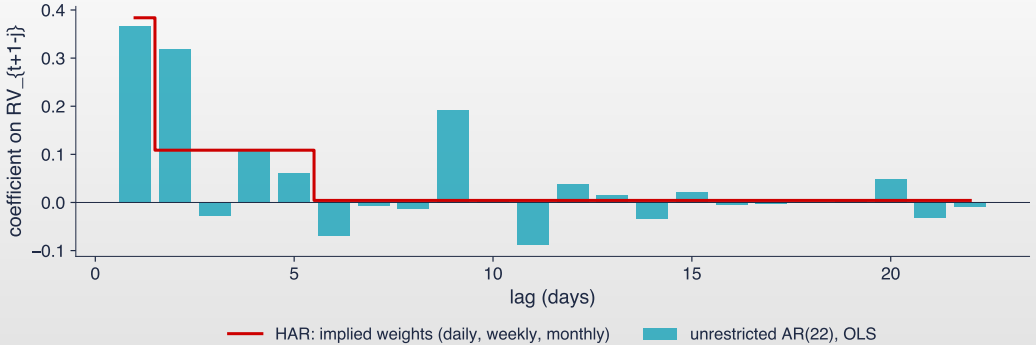
$$\mathrm{RV}_t^{(w)} = \frac{1}{5}\sum_{j=0}^{4} \mathrm{RV}_{t-j}, \qquad \mathrm{RV}_t^{(m)} = \frac{1}{22}\sum_{j=0}^{21} \mathrm{RV}_{t-j}$$

  - $\mathrm{RV}_t^{(w)}$, $\mathrm{RV}_t^{(m)}$: averages over the last 5 trading days (a week) and the last 22 (a month)
  - $\beta_d, \beta_w, \beta_m \geq 0$: weights of the three horizons; β_0: intercept; u_{t+1}: forecast error with mean zero
  - $\beta_d + \beta_w + \beta_m$: the persistence; the closer to 1, the slower volatility returns to its mean
- Heterogeneous market hypothesis: daily, weekly and monthly traders react to volatility at their own horizon
  - the cascade produces slowly decaying autocorrelations

The HAR model of Corsi (2/2): estimation and variants

- Statistically, HAR is an AR(22) with 19 linear restrictions (step-shaped coefficients)
  - short memory, but it mimics long memory over a month
- OLS is consistent; the errors are heteroskedastic and, for multi-day targets, overlapping
  - use Newey–West Newey and West (1987) (HAC) standard errors
- Variants
  - log-HAR: the same regression on $\ln \mathrm{RV}$, with near-Gaussian errors; the forecast of RV is $\exp(\hat{\mu} + \frac{1}{2}\hat{\sigma}_u^2)$, with $\hat{\mu}$ the fitted log value and $\hat{\sigma}_u^2$ the residual variance
  - HAR on $\sqrt{\mathrm{RV}}$; h-day targets (the average RV over the next h days)
- Extensions: continuous and jump parts, HAR-CJ Andersen, Bollerslev and Diebold (2007); positive and negative semivariances Patton and Sheppard (2015); implied volatility; leverage terms

HAR as a restricted AR(22)



- S&P 500 (Oxford-Man, 5-minute RV, 2000–2022, $T = 5\,530$): coefficients of an unrestricted AR(22) by OLS and the step weights implied by the HAR estimates

ATS_ch8_har

Interpreting the HAR estimates

- HAR: $\hat{\beta}_d = 0.275$ (0.104), $\hat{\beta}_w = 0.522$ (0.140), $\hat{\beta}_m = 0.094$ (0.078), Newey–West s.e.; $R^2 = 0.560$; persistence $\sum \hat{\beta} = 0.891$
- The AR(22) with 23 coefficients reaches $R^2 = 0.596$ in sample, with erratic coefficients; HAR captures the shape with 4
- Log-HAR: $\hat{\beta}_d = 0.393$, $\hat{\beta}_w = 0.399$, $\hat{\beta}_m = 0.153$, $R^2 = 0.719$ on $\ln \mathrm{RV}$; in logs the monthly component matters more
- In levels a few crisis days dominate OLS: large standard errors on β_d and β_m; weighted least squares or logs give more stable estimates

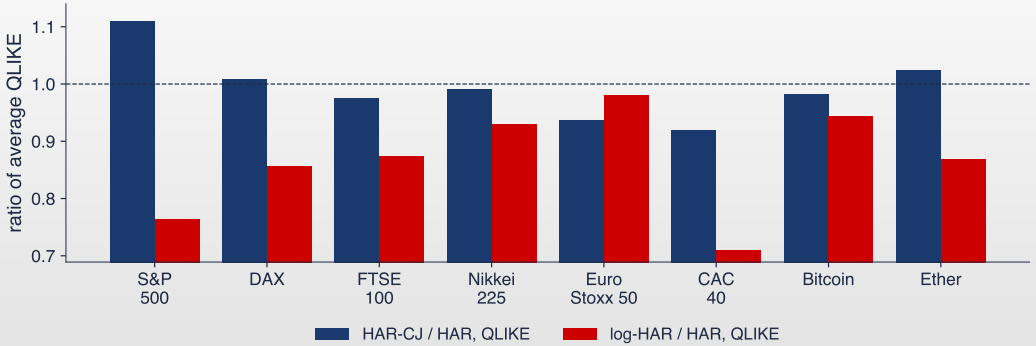
Forecasting design

- One day ahead, rolling window of 1 000 days, re-estimated every day; target RV_{t+1}
- Losses: MSE $= (\mathrm{RV} - f)^2$ and QLIKE, both robust to the noise in RV (see below)

$$\mathrm{QLIKE} = \frac{\mathrm{RV}}{f} - \ln \frac{\mathrm{RV}}{f} - 1 \geq 0$$

  - f: the variance forecast; QLIKE is 0 when $f = \mathrm{RV}$ and penalises under-prediction more than over-prediction
- Insanity filter Bollerslev, Patton and Quaedvlieg (2016): a forecast outside the range of the in-sample RV is replaced by the in-sample mean
  - linear HAR-type forecasts can be negative after a spike; QLIKE is then undefined
  - the filter is part of the method and must be reported
- Diebold–Mariano Diebold and Mariano (1995) t-statistic of the loss differential against HAR
  - Newey–West variance; negative = better than HAR, below -1.96 significant at 5%; several models: the model confidence set Hansen, Lunde and Nason (2011), Chapter 1
- Models: HAR, HAR-CJ (continuous part and jump part, from BV), log-HAR; for Bitcoin and Ether also HARQ

HAR extensions out of sample



- Average QLIKE relative to HAR (below 1 = better), six Oxford-Man indices (2004–2022) and two cryptocurrencies (from 01.11.2020); 1 000-day rolling window

ATS_ch8_har

Interpreting the out-of-sample HAR comparison

- Log-HAR beats HAR in QLIKE for 6 of 6 indices: S&P 500 0.764 (DM -2.07), CAC 40 0.709 (DM -1.73), FTSE 0.874 (DM -5.08)
- HAR-CJ helps in 4 of 6 indices, but not for the S&P 500 (1.110): separating jumps adds estimation noise when jumps are small
- The ranking depends on the loss: in MSE the gains of log-HAR are smaller (S&P 500: 0.837); MSE weighs the crisis days most
- Six indices from one data vendor and one period: report all of them, not the best one (data snooping, Chapter 1)

HARQ (1/2): the measurement error of RV

- RV is the latent IV plus a measurement error whose variance changes every day

$$\mathrm{RV}_t = \mathrm{IV}_t + e_t, \qquad \mathrm{Var}(e_t|\mathrm{IQ}_t) \approx \frac{2\,\mathrm{IQ}_t}{n}$$

  - e_t: measurement error of day t; on volatile days (IQ_t large) RV is less precise

- Errors in variables: in an AR(1) for IV with coefficient ϕ, the OLS slope on RV is attenuated (Appendix)

$$\operatorname{plim} \hat{\phi} = \phi\lambda, \qquad \lambda = \frac{\mathrm{Var}(\mathrm{IV})}{\mathrm{Var}(\mathrm{IV}) + \mathbb{E}\mathrm{Var}(e)} \in (0,1)$$

  - λ: reliability ratio, the share of the variance of RV that is signal
  - one constant β_d is a compromise: too high on noisy days, too low on precise days

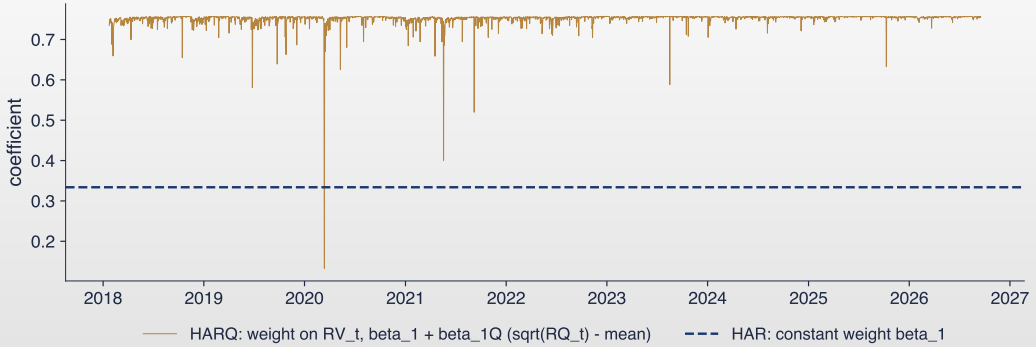
HARQ (2/2): a weight that depends on precision

- Bollerslev, Patton and Quaedvlieg (2016): the weight of yesterday's RV moves with its estimated precision

$$\mathrm{RV}_{t+1} = \beta_0 + \left(\beta_d + \beta_{dQ}\sqrt{\mathrm{RQ}_t}\right)\mathrm{RV}_t + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + u_{t+1}$$

  - RQ_t: realised quarticity, the estimate of IQ_t; $\sqrt{\mathrm{RQ}_t}$ is proportional to the standard deviation of e_t
  - $\beta_{dQ} < 0$ expected: the less precise yesterday's RV, the lower its weight
  - $\sqrt{\mathrm{RQ}_t}$ is demeaned, so β_d is the weight on a day of average precision
- One extra parameter, still OLS
- The original paper: S&P 500 futures and 27 Dow Jones stocks, gains over HAR in and out of sample
  - our replication extends it to Bitcoin and Ether, 2018–2026, where RQ is available from our one-minute data

HARQ for Bitcoin: a weight that moves with precision



- Full-sample HARQ on Bitcoin 5-minute RV and RQ: daily weight $\hat{\beta}_d + \hat{\beta}_{dQ}(\sqrt{\mathrm{RQ}_t} - \overline{\sqrt{\mathrm{RQ}}})$ against the constant HAR weight

ATS_ch8_har

Interpreting HARQ

- $\hat{\beta}_{dQ} = -0.00021$ ($t = -2.7$, Newey–West): negative, as predicted; the daily weight at average precision is 0.752, against 0.334 in HAR
- On most days the weight is near 0.76; on the noisiest days it falls (1% quantile 0.71, minimum 0.13): yesterday's RV is trusted only when it is precise
- Out of sample, QLIKE relative to HAR: Bitcoin 0.946 (DM -2.44), Ether 0.921 (DM -4.50): the gain of the original study carries over to crypto
- In MSE the picture differs (Bitcoin 1.163): MSE is dominated by a few extreme days, where HARQ shrinks the most; the AI mini-case checks how fragile the gain is

Realized GARCH (1/2): the three equations

- Hansen, Huang and Shek (2012), log-linear form: a return equation, a GARCH equation driven by the realised measure, and a measurement equation

$$r_t = \sqrt{h_t}\, z_t, \qquad \ln h_t = \omega + \beta \ln h_{t-\mathbf{1}} + \gamma \ln x_{t-\mathbf{1}}$$
$$\ln x_t = \xi + \varphi \ln h_t + \tau(z_t) + u_t$$

  - h_t: conditional variance of the return r_t; z_t: i.i.d. standardised shock; x_t: realised measure of day t (here the realised kernel RK)
  - β: persistence of $\ln h_t$; γ: weight of yesterday's realised measure (the news variable that replaces $\varepsilon^{\mathbf{2}}_{t-\mathbf{1}}$)
  - u_t: measurement error, i.i.d. $N(0, \sigma_u^{\mathbf{2}})$, independent of z_t
- Measurement equation: x_t is a noisy, possibly biased signal of h_t
  - $\varphi = 1$: x_t proportional to h_t; ξ absorbs the scale (open-to-close measure against close-to-close variance)
  - leverage function $\tau(z) = \tau_{\mathbf{1}} z + \tau_{\mathbf{2}}(z^{\mathbf{2}} - 1)$: with $\tau_{\mathbf{1}} < 0$, negative returns raise the next realised measure (news impact, TSA, Chapter 5)

Realized GARCH (2/2): reduced form and estimation

- Substituting the measurement equation into the GARCH equation: $\ln h_t$ is an AR(1)

$$\ln h_t = (\omega + \gamma\xi) + \pi \ln h_{t-1} + \gamma\big(\tau(z_{t-1}) + u_{t-1}\big), \qquad \pi = \beta + \varphi\gamma$$

  - π: persistence of the log variance; multi-step forecasts follow, because x_t has its own equation
- Joint QML: the log-likelihood sums a return part and a measurement part

$$\ell = -\frac{1}{2}\sum_t \Big[\underbrace{\ln h_t + z_t^2}_{r_t} + \underbrace{\ln \sigma_u^2 + u_t^2/\sigma_u^2}_{x_t}\Big]$$

  - comparison with GARCH only through the partial likelihood of r_t (the first part), since GARCH does not model x_t

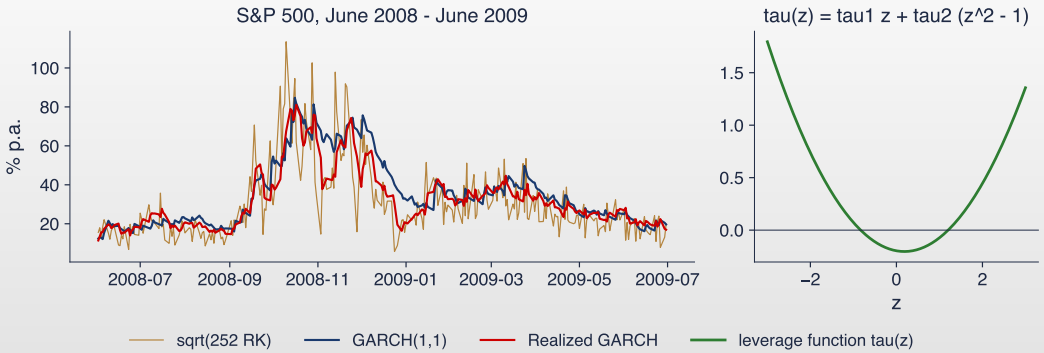
HEAVY and the family of realised-measure models

- HEAVY Shephard and Sheppard (2010): two GARCH-type equations, one for the return variance and one for the mean of the realised measure

$$h_t = \mathrm{Var}(r_t|\mathcal{F}_{t-\mathbf{1}}) = \omega + \alpha \mathrm{RM}_{t-\mathbf{1}} + \beta h_{t-\mathbf{1}}$$
$$\mu_t = \mathbb{E}(\mathrm{RM}_t|\mathcal{F}_{t-\mathbf{1}}) = \omega_R + \alpha_R \mathrm{RM}_{t-\mathbf{1}} + \beta_R \mu_{t-\mathbf{1}}$$

  - RM_t: realised measure of day t; (ω, α, β) and $(\omega_R, \alpha_R, \beta_R)$: the parameters of the two equations
  - each equation is estimated separately by QML; momentum: after a shock, h_t keeps rising while μ_t catches up
  - when the realised measure is in the variance equation, the squared return usually adds nothing ($\varepsilon^{\mathbf{2}}_{t-\mathbf{1}}$ gets a zero coefficient)
- Realized EGARCH Hansen and Huang (2016): several realised measures and a richer leverage; score-driven versions exist for fat tails
- Compared with HAR: returns and realised measures are modelled jointly, so densities and VaR/ES forecasts are available (Chapter 9)

Case study: Realized GARCH for the S&P 500



- Open-to-close returns and the Parzen realised kernel (Oxford-Man), 2000–2022, $T = 5\,552$, as in Hansen, Huang and Shek (2012); left: the 2008 crisis; right: the estimated leverage function

ATS_ch8_realized_garch

Realized GARCH estimates

ω	β	γ	ξ	φ	τ_1	τ_2	σ_u
0.119	0.678	0.287	−0.456	1.012	−0.073	0.197	0.671
(0.010)	(0.018)	(0.016)	(0.021)	(0.026)	(0.008)	(0.010)	(0.008)

- Bollerslev–Wooldridge s.e. in brackets; persistence $\hat{\pi} = \hat{\beta} + \hat{\varphi}\hat{\gamma} = 0.968$ (GARCH(1,1) on the same returns: 0.989)
- Partial log-likelihood of the returns: Realized GARCH −6899.7, HEAVY −6902.8, GJR −6998.8, GARCH(1,1) −7099.0
- HEAVY variance equation: $\hat{\alpha} = 0.377$ on RK_{t-1}, $\hat{\beta} = 0.691$

Interpreting Realized GARCH

- $\hat{\varphi} = 1.012$: the kernel is proportional to the conditional variance; $\hat{\xi} = -0.456$ with $\hat{\sigma}_u = 0.671$: on average the kernel lies below the conditional variance of the open-to-close return
- $\hat{\gamma} = 0.287$ is large and $\hat{\beta} = 0.678$ smaller than in GARCH: the realised measure brings information faster than squared returns (left panel, autumn 2008)
- Leverage: $\hat{\tau}_1 = -0.073$, $\hat{\tau}_2 = 0.197$: an asymmetric parabola, larger for negative z
- The gain in the partial likelihood over GARCH(1,1) (199.3 points) has the direction of the original paper; HEAVY is almost as good with three parameters

The proxy problem and robust losses (1/2)

- We never see σ_t^2: a forecast h_t is compared with a proxy $\hat{\sigma}_t^2$ (squared return, RV, RK)
  - the proxy is conditionally unbiased, $\mathbb{E}(\hat{\sigma}_t^2 | \mathcal{F}_{t-1}) = \sigma_t^2$, but noisy
- A loss L is *robust* Patton (2011) if the ranking of forecasts by $\mathbb{E}L(\hat{\sigma}_t^2, h_t)$ equals the ranking by $\mathbb{E}L(\sigma_t^2, h_t)$, for every unbiased proxy Hansen and Lunde (2006a)
- Necessary and sufficient condition: a Bregman form (Chapter 1, consistent scoring functions for the mean)

$$L(\hat{\sigma}^2, h) = \tilde{C}(h) + B(\hat{\sigma}^2) + C(h)(\hat{\sigma}^2 - h), \qquad C'(h) < 0$$

  - C: a decreasing function of the forecast; $\tilde{C}$: its antiderivative, $\tilde{C}' = C$; B: any function of the proxy alone (it does not affect the ranking)

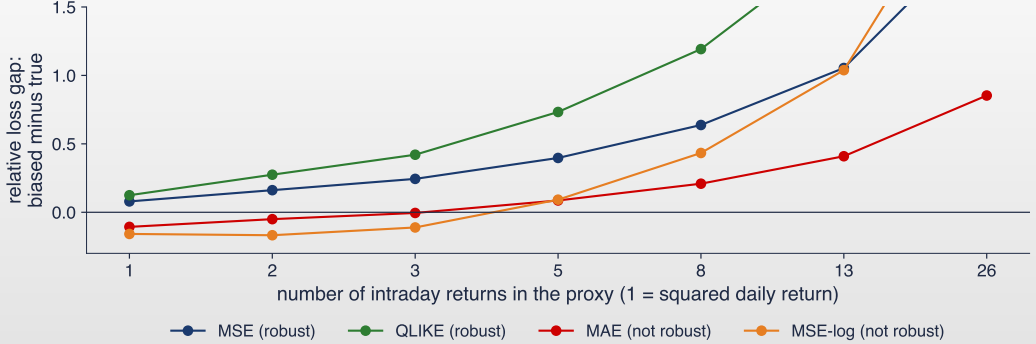
The proxy problem and robust losses (2/2)

- Robust: MSE and QLIKE, the members of degree 2 and degree 0 of the homogeneous robust family

$$\mathrm{MSE} = (\hat{\sigma}^2 - h)^2, \qquad \mathrm{QLIKE} = \frac{\hat{\sigma}^2}{h} - \ln\frac{\hat{\sigma}^2}{h} - 1$$

  - degree: how the loss scales when proxy and forecast are multiplied by the same constant; QLIKE (degree 0) is scale-free and penalises under-prediction more
- Not robust: MAE $= |\hat{\sigma}^2 - h|$, MSE on logs $(\ln\hat{\sigma}^2 - \ln h)^2$, MSE on standard deviations
  - with a noisy proxy they reward forecasts that are biased downwards
- Example: with $\hat{\sigma}^2 = r^2$, the MSE-log optimum is $h^* = \exp(\mathbb{E}\ln r^2) = 0.28\,\sigma^2$
  - because $r^2 = \sigma^2\chi_1^2$ for a Gaussian return and $\mathbb{E}\ln\chi_1^2 = -1.27$: MSE-log prefers a forecast far below the truth

Robust and non-robust losses



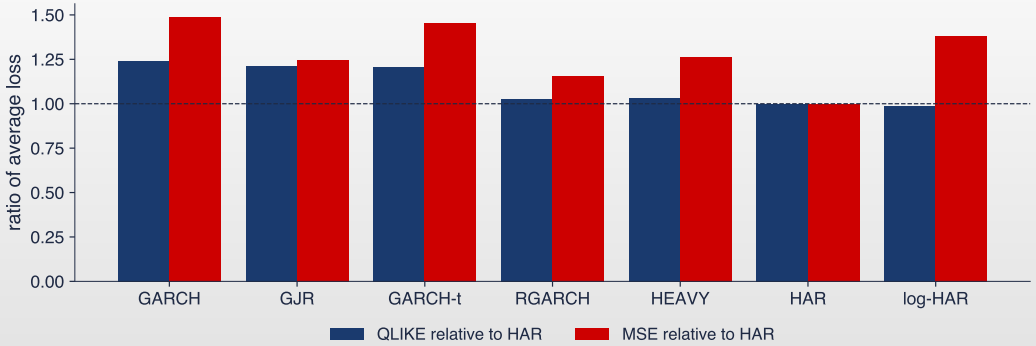
- True variance against a forecast biased down by the factor 0.6; proxy = RV from n intraday returns ($n = 1$: squared daily return); relative gap of expected losses, above zero = the true variance wins

ATS_ch8_robust_loss

Interpreting the robustness experiment

- With the squared return as proxy, MAE prefers the biased forecast (gap -10.7%) and so does MSE-log (-15.8%); MSE (7.9%) and QLIKE (12.5%) do not
- With $n = 5$ intraday returns all four losses rank correctly: a precise proxy makes non-robust losses "almost" robust Hansen and Lunde (2006a)
- The danger is largest exactly where realised measures are unavailable: daily data, emerging markets, long histories
- Rule: evaluate volatility forecasts with QLIKE (and MSE), against the best available proxy

Seven forecasts of S&P 500 volatility



- Open-to-close variance, one day ahead, 2016–2022 (1 537 days); expanding window from 2000, parameters re-estimated every 250 days; proxy: realised kernel

ATS_ch8_robust_loss

Interpreting the forecast comparison

- QLIKE: HAR 0.328, log-HAR 0.324, Realized GARCH 0.337, HEAVY 0.339, GARCH-t 0.396, GJR 0.398, GARCH 0.407
- DM against HAR (QLIKE): GARCH 8.62, GJR 6.27, Realized GARCH 1.42, log-HAR -0.37: realised-measure models beat return-only models clearly; among them the differences are small
- In MSE, HAR (1.60) leads and the DM statistics are smaller: the 2020 crash dominates the squared errors
- One proxy (RK) for all models: the HAR family is fitted to that same measure, which favours it; a fair comparison also tries another proxy (RV, BV)

From one variance to a covariance matrix (1/2)

- The vector of N return shocks is a matrix square root of the conditional covariance times a standardised vector

$$\varepsilon_t = H_t^{1/2}\eta_t, \qquad \eta_t \text{ i.i.d. } (0, I_N)$$

  - $H_t = \mathrm{Var}(\varepsilon_t|\mathcal{F}_{t-1})$: $N \times N$ conditional covariance matrix; I_N: identity matrix
  - H_t must be symmetric positive definite for every t and every parameter value
- VEC: every element of H_t depends on all past squares and cross-products

$$\mathrm{vech}(H_t) = c + A\,\mathrm{vech}(\varepsilon_{t-1}\varepsilon_{t-1}') + B\,\mathrm{vech}(H_{t-1})$$

  - vech: stacks the $N(N+1)/2$ distinct elements of a symmetric matrix; c: vector, A, B: square matrices of that size
  - general, but positivity is hard to impose

From one variance to a covariance matrix (2/2): BEKK, CCC and DCC

- BEKK Engle and Kroner (1995): quadratic forms guarantee positive definiteness

$$H_t = CC' + A'\varepsilon_{t-1}\varepsilon_{t-1}'A + B'H_{t-1}B$$

  - C: lower-triangular $N \times N$; A, B: $N \times N$ matrices; diagonal and scalar versions restrict A, B
  - identification: (A, B) and $(-A, -B)$ give the same H_t; fix the sign of one element

- CCC Bollerslev (1990) and DCC Engle (2002): variances from univariate GARCH, correlations modelled separately

$$H_t = D_t R D_t \ \text{(CCC)}, \qquad H_t = D_t R_t D_t \ \text{(DCC)}, \qquad D_t = \operatorname{diag}(\sigma_{1t}, \ldots, \sigma_{Nt})$$

  - σ_{it}: conditional standard deviation of asset i from its own GARCH; R: constant correlation matrix; R_t: time-varying correlation matrix

The curse of dimensionality in numbers

N	VEC	BEKK	diagonal BEKK	scalar BEKK	DCC
2	21	11	7	5	9
5	465	65	25	17	27
10	6 105	255	75	57	77
25	211 575	1 575	375	327	377
50	3 252 525	6 275	1 375	1 277	1 377
100	51 010 050	25 050	5 250	5 052	5 252

- Counts of free parameters of the (1,1) versions; DCC: $3N$ univariate GARCH parameters, 2 correlation parameters and the $N(N-1)/2$ elements of the target
- The DCC target is estimated by moments (correlation targeting), so the numerical search has only 2 parameters at any N; the price: $N(N-1)/2$ estimated correlations

Asymptotic theory for multivariate GARCH

- Gaussian QML is the standard estimator; consistency and asymptotic normality: VEC and BEKK Comte and Lieberman (2003), Hafner and Preminger (2009); CCC with ARMA means Ling and McAleer (2003)
- Conditions are stronger than in the univariate case (moments of order 6 or 8 for BEKK in early results); the sandwich of the univariate section carries over
- DCC is estimated in two steps Engle and Sheppard (2001): (1) univariate GARCH for each series; (2) QML of the correlation part given the standardised residuals $z_t = D_t^{-1}\varepsilon_t$
  - second-step standard errors must account for the first step (a GMM-type correction); naive second-step s.e. are too small
- Large N: the full likelihood needs R_t^{-1} and $|R_t|$ at each t ($O(N^3)$) and the score is dominated by noise; composite likelihood over pairs Pakel, Shephard, Sheppard and Engle (2021) avoids both

DCC and its corrected version cDCC (1/2): DCC



Robert F. Engle, 2017

Photo: Econterms (2017); CC BY-SA 4.0; Wikimedia Commons

- DCC: a GARCH(1,1)-type recursion for a quasi-correlation matrix Q_t, rescaled into a correlation matrix

$$Q_t = (1 - a - b)S + a\, z_{t-1} z_{t-1}' + b\, Q_{t-1}$$

$$R_t = \mathrm{diag}(Q_t)^{-1/2}\, Q_t\, \mathrm{diag}(Q_t)^{-1/2}$$

  - $z_t = D_t^{-1}\varepsilon_t$: standardised residuals of the univariate GARCH models; S: their sample correlation matrix (the target)
  - $a \geq 0$: reaction of correlations to yesterday's co-movement; $b \geq 0$: persistence; $a + b < 1$
  - $\mathrm{diag}(Q_t)$: the diagonal matrix of the diagonal elements of Q_t; the rescaling puts ones on the diagonal of R_t

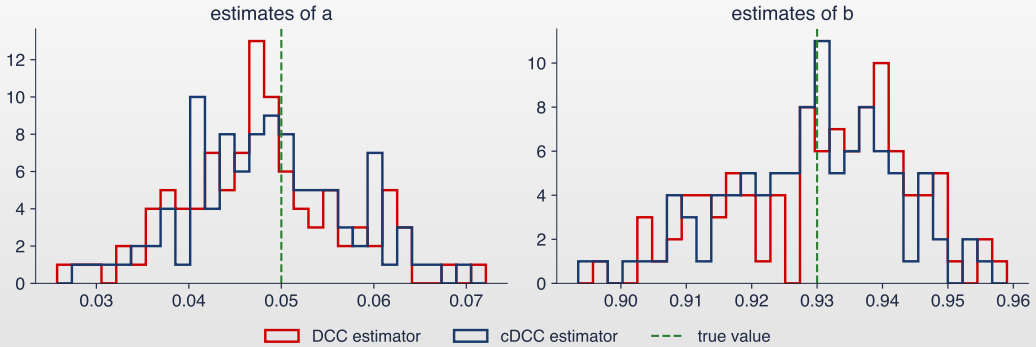
DCC and its corrected version cDCC (2/2): cDCC

- Aielli (2013): the DCC target is estimated inconsistently
  - $\mathbb{E}(z_t z_t' | \mathcal{F}_{t-1}) = R_t \neq Q_t$, so in general $\mathbb{E}Q_t \neq \mathbb{E}z_t z_t'$
  - the moment estimator of S is inconsistent, and so is $(\hat{a}, \hat{b})$
- cDCC: rescale the residuals by the diagonal of Q_{t-1} before they enter the recursion

$$z^*_{t-1} = \mathrm{diag}(Q_{t-1})^{1/2} z_{t-1}, \qquad Q_t = (1 - a - b)S + a\, z^*_{t-1} z^{*\prime}_{t-1} + b\, Q_{t-1}$$

  - then $\mathbb{E}(z^*_t z^{*\prime}_t | \mathcal{F}_{t-1}) = Q_t$, and $S = \mathbb{E}z^*_t z^{*\prime}_t$ is a valid target
  - the cDCC target depends on (a, b): it is estimated iteratively inside the likelihood

DCC against cDCC: a simulation



- Bivariate cDCC process with $a = 0.05$, $b = 0.93$, target correlation 0.5, unit variances; $T = 2\,000$, 100 replications; both estimators on each sample

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Interpreting the DCC simulation

- Mean estimates: DCC $\hat{a} = 0.0478$, $\hat{b} = 0.9307$; cDCC $\hat{a} = 0.0486$, $\hat{b} = 0.9288$ (true 0.05 and 0.93)
- RMSE of $\hat{a}$: 0.0090 (DCC) and 0.0086 (cDCC); of $\hat{b}$: 0.0135 and 0.0129: the inconsistency is real but small at these persistence levels
- Sampling error dominates the bias with two thousand observations; the difference matters more for the target in large systems and for inference on (a, b)
- A proof of inconsistency does not tell you the size of the bias: simulate it in your own design before choosing an estimator

Large covariance matrices: shrinkage and DCC-NL (1/2)

- With N/T not small, the eigenvalues of a sample covariance S are too dispersed
  - N: number of assets; T: number of observations; the largest eigenvalues are too large, the smallest too small; the inverse S^{-1} amplifies the error
- Global minimum-variance (GMV) portfolio: the weights with the smallest variance that sum to 1

$$w = \frac{\Sigma^{-1}1}{1'\Sigma^{-1}1}$$

  - Σ: covariance matrix of the returns; 1: vector of ones
  - it loads on the directions with the smallest, most underestimated eigenvalues
  - it depends only on Σ, so its realised risk measures the quality of the covariance forecast without noise from expected returns

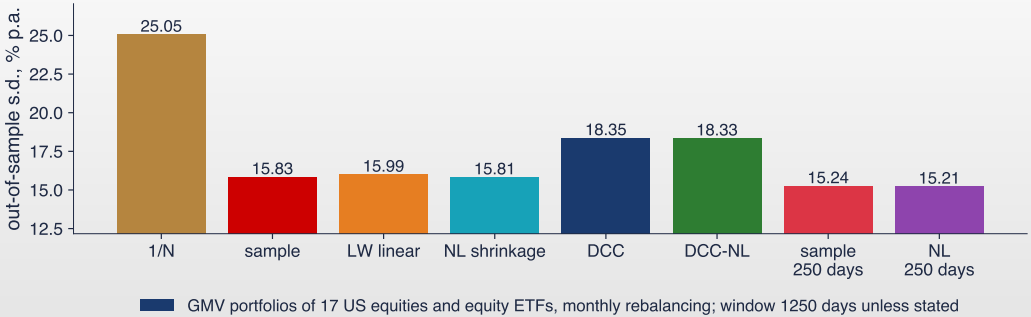
Large covariance matrices: shrinkage and DCC-NL (2/2)

- Linear shrinkage Ledoit and Wolf (2004): a weighted average of S and a scaled identity

$$\hat{\Sigma} = \delta\,\mu I + (1-\delta)\,S$$

  - μ: average eigenvalue of S; $\delta \in [0,1]$: shrinkage intensity, estimated from the data (larger when N/T is larger)
- Nonlinear shrinkage Ledoit and Wolf (2020): keep the eigenvectors of S, replace each eigenvalue λ_i by $d(\lambda_i)$
  - $d(\cdot)$: an analytical kernel formula that pulls small eigenvalues up and large ones down, by different amounts
- DCC-NL Engle, Ledoit and Wolf (2019): in DCC, the target S of the standardised residuals is itself a large sample covariance: replace it by its nonlinear shrinkage
  - composite likelihood for (a, b), univariate GARCH for the variances; evaluated by the out-of-sample s.d. of GMV portfolios

Case study: Engle, Ledoit and Wolf (2019) on US equities



- 17 assets: 14 US stocks and three equity ETFs (SPY, QQQ, RSP), which are portfolios of stocks; GMV weights rebalanced every 21 days from 23.05.2017 (112 rebalancings); window 1 250 days, and 250 days for the last two bars; annualised out-of-sample s.d.

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Interpreting the GMV comparison (1/2)

- Out-of-sample s.d. (% p.a.): 1/N 25.05, sample 15.83, linear shrinkage 15.99, nonlinear shrinkage 15.81, DCC 18.35, DCC-NL 18.33
- With $N/T = 17/1\,250$ the sample covariance is already accurate
  - nonlinear shrinkage changes little (15.81 against 15.83); DCC-NL is no better than DCC
- Here the dynamics hurt: DCC gives 18.35 against 15.83
  - DCC uses GARCH variances, with forecasts averaged over the 21-day holding period
  - one-month-ahead variance forecasts of single stocks such as MSTR and TSLA are noisy, and GMV amplifies their errors

Interpreting the GMV comparison (2/2)

- A window of 250 days makes N/T five times larger
  - s.d. of the sample GMV 15.24, with nonlinear shrinkage 15.21: only 0.2% lower
  - yet the ETFs make the correlation matrix nearly singular (median condition number 1146)
  - the shorter window itself lowers the risk: a crude form of time variation
- The original study uses hundreds of stocks, where N/T is large
  - there the DCC-NL gains are substantial
  - with our N the lesson is when shrinkage matters, not how much it gains

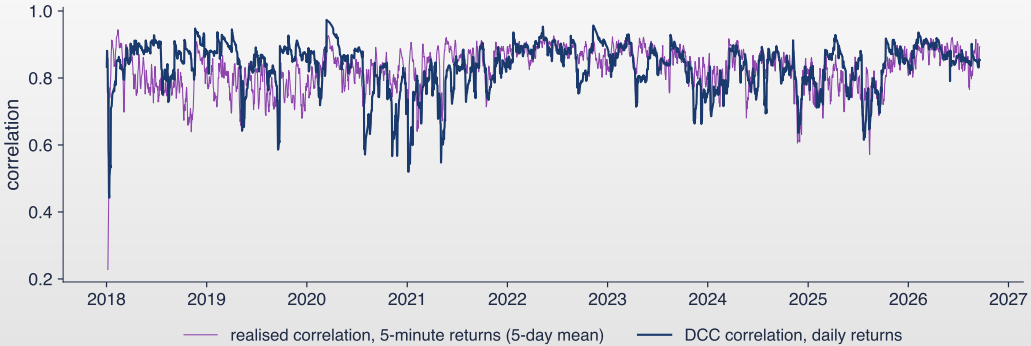
Realised covariance

- The multivariate RV: the sum of outer products of the intraday return vectors

$$\mathrm{RCov}_t = \sum_{i=1}^{n} r_{t,i}\, r_{t,i}' \to \int_0^1 \Sigma_s \, ds$$

  - $r_{t,i}$: $N \times 1$ vector of intraday returns; Σ_s: spot covariance matrix; the limit adds the co-jumps (simultaneous jumps); with synchronous, noise-free prices it inherits the CLT of RV
- Asynchronous trading: previous-tick prices create zero returns for the asset that has not traded; covariances shrink towards zero as the interval falls (the Epps effect)
  - refresh-time sampling (all assets have traded) and the multivariate realised kernel Barndorff-Nielsen et al. (2011): consistent and positive semi-definite
- Forecasting: HAR on the elements of the Cholesky factor of RCov_t keeps forecasts positive definite Chiriac and Voev (2011); HEAVY and Realized GARCH have multivariate versions
- Evaluating covariance forecasts H_t against a matrix proxy Σ_t: robust losses Laurent, Rombouts and Violante (2012)
  - matrix QLIKE $\ln|H_t| + \mathrm{tr}(H_t^{-1}\Sigma_t)$, with $|\cdot|$ the determinant and tr the trace; Frobenius MSE, the sum of squared element-wise errors

Bitcoin and Ether: realised and DCC correlation



- Daily realised correlation from 5-minute returns (5-day means) and the DCC correlation of daily close-to-close (UTC) returns, 2018–2026

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Interpreting the correlations

- Average realised correlation 0.83, average DCC correlation 0.83; correlation between the two series 0.26
- DCC uses one observation a day ($\hat{a} = 0.051$, $\hat{b} = 0.928$): it smooths and reacts with a lag; the realised measure moves daily and drops quickly in idiosyncratic episodes (5% quantile 0.68)
- Combining both (a realised measure in the correlation equation) is the multivariate analogue of Realized GARCH
- Five-minute sampling sits on the flat part of the Epps curve for these two coins (signature slide): the realised correlation is not biased towards zero

An open question

- Does exploiting the measurement error of realised variance (HARQ) improve volatility forecasts robustly, or only under particular design choices?
  - formal: the QLIKE ratio HARQ/HAR and its DM statistic across a pre-registered grid (window, filter, realised measure, asset); falsified if the ratio exceeds 1 or the DM test is insignificant for most cells
  - the measure of precision (RQ) is itself noisy and fat-tailed, so the gain may be fragile
- Why it matters: risk systems use one-day volatility forecasts every day; a small but robust gain is worth more than a large fragile one
  - literature to start from: Bollerslev, Patton and Quaedvlieg (2016), Corsi (2009), Patton (2011), Hansen and Lunde (2006a), Hansen, Lunde and Nason (2011)

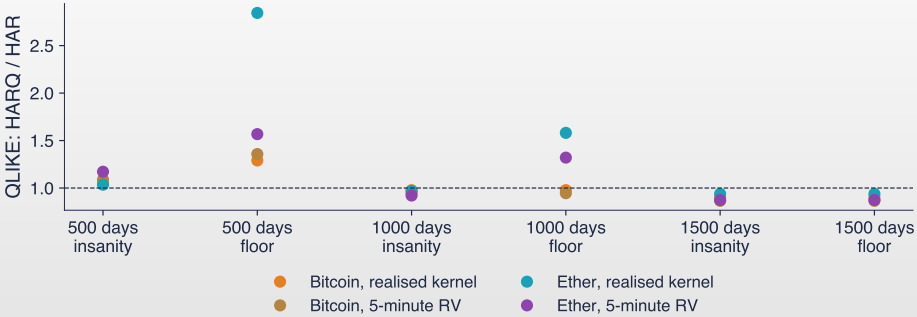
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List peer-reviewed papers since 2016 that test HARQ-type models out of sample on assets other than US equities; give DOIs.` Then check every DOI on Crossref
  - **hypothesis**: `Under which data features (24/7 trading, jumps, zero returns) should the HARQ gain shrink or grow? Give testable predictions.`
  - **code and replication**: ask for HAR and HARQ rolling forecasts, then reproduce a known number first (the Bitcoin QLIKE ratio of this lecture)
  - **robustness and critique**: `Act as a hostile referee: which choices in a HARQ forecast comparison could produce a spurious gain?`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- The forecast at t uses only data up to t: windows, demeaning of RQ, the insanity filter and the bias correction of log forecasts are all computed in the estimation window
- The loss is robust (QLIKE, MSE) and the proxy is stated; DM statistics use HAC variances
- All cells of the grid are reported, not the best one; the number of comparisons is stated
- Realised measures are recomputed from raw prices and agree with an independent source on overlapping assets and dates

Mini-case: robustness of the HARQ–HAR comparison



- QLIKE ratio HARQ/HAR for Bitcoin and Ether: window 500, 1 000 or 1 500 days; insanity filter or a floor at the smallest in-sample RV; target 5-minute RV or realised kernel
- Ratios from 0.865 to 2.844; HARQ better in 58% of the 24 cells, significantly better (DM < -1.96) in 50%, significantly worse in 12%: an AI summary reporting one cell would hide most of this

ATS_ch8_ai_robustness

Project idea

- **Realised volatility of crypto-assets: does the equity toolkit transfer?**: replicate first, then extend
  - replicate: HAR and HARQ of this lecture for Bitcoin (QLIKE ratio 0.946) and Realized GARCH for the S&P 500 (persistence 0.968)
  - extend: ten coins from the Binance archive; noise-robust measures from one-second data
  - extend the tests: an intraday seasonality correction before jump tests; Realized GARCH with the realised kernel
  - use the spot-ETF date (January 2024) as a natural break
  - pre-register: assets, sample, measures, models, windows, losses, the DM/MCS tests and the robustness grid
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- Gaussian QML needs only the variance equation; with fat tails, use sandwich standard errors
- Volatility has fast and slow components; macro drivers of the slow one need long samples and honest variance ratios
- Realised measures make volatility observable with a known error; noise and jumps decide the estimator
- HAR, HARQ and Realized GARCH beat return-only models one day ahead; evaluate with QLIKE or MSE
- In many dimensions the correlation target needs shrinkage (DCC-NL); whether the dynamics help must be checked out of sample

Self-assessment

Questions

- Why are Hessian standard errors too small for GARCH when $\kappa_\eta > 3$?
- What does the variance ratio of GARCH-MIDAS measure?
- Why does stable convergence matter for the CLT of RV?
- Why can a signature plot rise at high frequency?
- Why is β_{dQ} expected to be negative in HARQ?
- Which losses keep the ranking under a noisy proxy?

Next:

- VaR, ES and backtesting: elicitability, scoring and model risk
- From volatility forecasts to tail forecasts: VaR 1% and ES 2.5%, their scoring functions and backtests

Appendix: the GARCH sandwich

- $\ell_t = -\frac{1}{2}(\ln \sigma_t^2 + \varepsilon_t^2/\sigma_t^2)$, $d_t = \sigma_t^{-2}\partial\sigma_t^2/\partial\theta$: $s_t = \frac{1}{2}(\eta_t^2 - 1)d_t$ at θ_0
- $B = \mathbb{E}s_t s_t' = \frac{1}{4}\mathbb{E}(\eta_t^2 - 1)^2\,\mathbb{E}d_t d_t' = \frac{\kappa_\eta - 1}{4}J$, because η_t is independent of $d_t \in \mathcal{F}_{t-1}$
- $\partial s_t/\partial\theta' = -\frac{1}{2}\eta_t^2 d_t d_t' + \frac{1}{2}(\eta_t^2 - 1)\partial d_t/\partial\theta'$; the second term has mean zero, so $A = \frac{1}{2}J$
- $A^{-1}BA^{-1} = 4J^{-1} \cdot \frac{\kappa_\eta - 1}{4}J \cdot J^{-1} = (\kappa_\eta - 1)J^{-1}$; Hessian only: $A^{-1} = 2J^{-1}$; OPG: $B^{-1} = \frac{4}{\kappa_\eta - 1}J^{-1}$

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Appendix: the noise bias of realised variance

- $\tilde{r}_i = Y_{i/n} - Y_{(i-1)/n} = r_i + u_i - u_{i-1}$, with u i.i.d. $(0, \omega^2)$ independent of X
- $\sum_i \tilde{r}_i^2 = \sum_i r_i^2 + 2\sum_i r_i(u_i - u_{i-1}) + \sum_i (u_i - u_{i-1})^2$; the cross term has mean zero, the last has mean $2n\omega^2$
- $\mathrm{Cov}(\tilde{r}_i, \tilde{r}_{i-1}) = -\omega^2$: first-order negative autocorrelation; the realised kernel adds $2\gamma_1 \approx -2n\omega^2$ back and removes the bias
- Positively autocorrelated returns (gradual price adjustment, stale prices) give $\gamma_1 > 0$ and a signature plot that rises with the interval; the kernel corrects both signs

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Appendix: attenuation and HARQ

- $\mathrm{IV}_{t+1} = c + \phi \mathrm{IV}_t + v_{t+1}$, observed $\mathrm{RV}_t = \mathrm{IV}_t + e_t$, e_t uncorrelated with IV_t and v_{t+1}
- OLS of RV_{t+1} on RV_t: $\operatorname{plim} \hat{\phi} = \frac{\mathsf{Cov}(\mathrm{IV}_{t+1}, \mathrm{IV}_t)}{\mathsf{Var}(\mathrm{IV}_t) + \mathsf{Var}(e_t)} = \phi\lambda$, $\lambda = \frac{\mathsf{Var}(\mathrm{IV})}{\mathsf{Var}(\mathrm{IV}) + \mathbb{E}\mathsf{Var}(e_t)}$
- The best linear predictor given today's precision uses $\phi\lambda_t$, $\lambda_t = \frac{\mathsf{Var}(\mathrm{IV})}{\mathsf{Var}(\mathrm{IV}) + 2\mathrm{IQ}_t/n}$: decreasing in IQ_t
- A first-order expansion in $\sqrt{\mathrm{IQ}_t}$ gives $\beta_d + \beta_{dQ}\sqrt{\mathrm{RQ}_t}$ with $\beta_{dQ} < 0$: the HARQ specification Bollerslev, Patton and Quaedvlieg (2016)

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